Check for updates

Global Policy

SPECIAL ISSUE EU Space Governance at the Threshold of a New Era

POLICY ANALYSIS OPEN ACCESS

EU Space Law and Earth's Boundaries: Integrating Environmental Impact Assessment and Corporate Due Diligence

Elena Cirkovic[1] | Vitali Braun[2]

[1]Aarhus University, Law Department, Aarhus, Denmark | [2]IMS Space Consultancy GmbH at ESA/ESOC, Darmstadt, Germany

Correspondence: Elena Cirkovic (elec8@law.au.dk)

Received: 23 January 2025 | **Revised:** 24 March 2025 | **Accepted:** 25 March 2025

Funding: The authors received no specific funding for this work.

Keywords: corporate social responsibility (CSR) | due diligence | Earth's planetary boundaries | EU space law

ABSTRACT

The European Union's proposed Space Law aims to establish a unified approach for sustainable space activities across the EU. While the EU's satellite constellations contribute to sustainability efforts, they can also have negative environmental impacts. This article argues that to address the challenges of space environment sustainability and the impact of space activities on Earth's planetary boundaries (EPB), the EU Space Law should incorporate environmental impact assessments (EIA) and public-private governance measures tailored for the space sector. The article examines potential interactions among EU space activities and existing legal frameworks, such as the outer space law regime, the EU's EIA Directive, and corporate social responsibility developments, including the Corporate Sustainability Due Diligence Directive (CSDDD). The analysis focuses on the relationship between the EPB approach and outer space activities, due diligence and strict liability in relation to intergenerational equity and future generations in international law, and the applicability of EIA and corporate governance to the emergent EU Space Law. The article emphasizes the importance of incorporating environmental and social considerations into the governance of space activities to protect outer space-Earth complex systems.

1 | European Union's Space Law Proposal and Earth's Planetary Boundaries

The European Union's forthcoming Space Law, announced by the European Commission in September 2023, aims to establish a unified approach across the EU for resilience, safety, and sustainability in space activities. By creating a standardized framework, the EU Space Law could drive socio-technical-environmental innovation and promote the sustainable use of space. The legislative process has involved stakeholder consultations and is expected to be finalized after the EU elections in June 2024 (Regulation (EU) 2023/588).

The context for this regulatory development is complex. While the existing EU satellite programmes contribute to the EU's self-sufficiency and support sustainability efforts (Regulation (EU) 2013 No 1285/2013, p1; Regulation (EU) 2014 No 377/2014, 44; Regulation (EU) 2010 No 911/2010, 1), the introduction of satellite constellations can also have negative environmental impacts, such as space debris, light and radiofrequency pollution, or atmospheric pollution (Miraux, Wilson, and Dominguez Calabuig 2022). As space flight activity increases, the space sector may become a contributor to various environmental impact categories, including climate change.

This article argues that in order to address the challenges of space environment sustainability and the impact of space activities on Earth's planetary boundaries (EPB), the EU Space Law could incorporate environmental impact assessments (EIA) and public-private governance measures tailored for

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2025 The Author(s). *Global Policy* published by Durham University and John Wiley & Sons Ltd.

entities operating in the space sector. This would involve developing a regulatory framework that mandates space actors to apply due diligence and adhere to best practices in corporate responsibility, environmental stewardship, and sustainable space operations.

To develop this analysis, we examine potential regime interactions among EU space activities and existing legal frameworks, such as the outer space law regime, the EU's EIA Directive, and ongoing developments and obstacles in corporate social responsibility.

The article is structured in three sections. The first section outlines the relationship between the EPB approach and outer space activities. The second section provides an overview of due diligence and strict liability in relation to intergenerational equity and future generations in international law. The third section considers recent developments in corporate governance in EU law and the applicability of EIA to the emergent EU Space Law.

2 | Earth Planetary Boundaries and Outer Space Activities

2.1 | Earth Planetary Boundaries Framework and Space Activities

Space technologies, particularly earth observation (EO) satellites, have become increasingly vital for monitoring global environmental issues, such as deforestation and ice melt, which are critical for managing climate change (Cirkovic 2025). A full life-cycle approach to space technologies is necessary, given the increased research, evidence, and awareness of its impact on the earth system environment (ESA-ESTEC 2024). Despite the indispensable role of EO in monitoring the Earth's system and climate change, most of the satellite sector is geared toward commercial telecommunications. According to ESA, 'the commercial telecommunications satellite manufacturing sector is a major part of the European space industry, representing around two-thirds of the €8.7 billion total revenues in 2020.' (ESA 2024).

While Earth planetary boundaries (EPB) studies might appear unrelated to research on space technology and sustainability, they demonstrate how human understanding and definitions of EPBs are constantly expanding. Moreover, they are examples of earth system processes that are or could become increasingly affected through the acceleration of increasing space activities.

Humanity's impact has surpassed planetary boundaries pertaining to environmental pollutants and other 'novel entities,' including plastics (Persson et al. 2022). Six of the nine boundaries are transgressed, suggesting that Earth is now well outside of the safe operating space for humanity. Ocean acidification is close to being breached, while aerosol loading regionally exceeds the boundary (Richardson et al. 2023).

In reference to recent studies on life cycles of space technology, outer space activities can affect existing nine EPBs as follows: climate change, atmospheric CO_2 concentration (parts per million, ppm) and, importantly, other greenhouse gas (GHG) emissions such as methane (CH_4); ocean acidification, changes in ocean pH; stratospheric ozone depletion, the concentration of ozone-depleting substances (e.g., CFCs); nitrogen cycle, human-induced perturbations to the global nitrogen cycle; phosphorus cycle: human-induced changes in the phosphorus cycle; global freshwater use, the ratio of human freshwater consumption to available renewable water resources; land use change, the percentage of the Earth's land area transformed by human activities; biodiversity loss, the rate of species extinction relative to the natural background rate; and aerosol loading, concentration of anthropogenic (human-caused) aerosols in the atmosphere (Miraux, Wilson, and Dominguez Calabuig 2022).

Recent research has identified significant connections between space activities and atmospheric composition. Murphy et al. (2023) found that approximately 10% of stratospheric sulfuric acid particles larger than 120 nm in diameter contain aluminum and other metals directly attributable to spacecraft reentry processes. Through single-particle mass spectrometry of over 500,000 aerosol particles, they identified more than 20 elements originating from spacecraft reentry, including lithium, aluminum, copper, niobium, silver, hafnium, and lead. For several elements—specifically lithium, aluminum, copper, and lead—the mass introduced through spacecraft reentry now exceeds the natural influx from cosmic dust. With projections of tens of thousands of additional satellites by 2030, up to half of all stratospheric sulfuric acid particles could contain metals from spacecraft reentry within decades, potentially affecting ice nucleation, aerosol layer size distribution, and atmospheric chemistry.

Concurrently, Parker et al. (2025) documented a reverse relationship: anthropogenic greenhouse gases cause cooling and contraction of the thermosphere, resulting in reduced atmospheric density where most low Earth orbit satellites operate. This density reduction has direct consequences for orbital debris management, as objects experience less atmospheric drag, remain in orbit longer, and pose increased collision hazards to operational satellites. The researchers utilized CO_2 emission projections to model how greenhouse gas emissions affect the "satellite carrying capacity" of low Earth orbit, establishing that greenhouse gas-induced thermospheric contraction extends orbital debris lifetime.

These studies collectively demonstrate a bidirectional relationship between Earth's atmosphere and orbital activities. More satellites lead to more reentry events, altering atmospheric composition, while greenhouse gas emissions extend debris lifetime in orbit, increasing collision risks and potentially generating more debris that will eventually reenter and further alter atmospheric composition. This feedback loop challenges the traditional separation between Earth's atmospheric management and space governance.

A planetary boundary approach would recognize the following:

1. Space activities have a complete lifecycle that includes atmospheric impacts during launch, operation, and reentry phases.
2. Earth's climate policies directly affect orbital sustainability

17585899, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1758-5899.70020 by New Aarhus University, Wiley Online Library on [27/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3. The atmosphere–orbit relationship requires consideration of cumulative effects and potential tipping points.
4. Current sectoral approaches to regulation fail to address these interconnected systems.

The technical challenges of space debris and spacecraft disposal illustrate a broader concern: the increasing interaction between space activities and Earth's environmental systems. While space debris poses immediate operational risks, the environmental impacts extend far beyond orbital space.

The surge in human-made materials entering Earth's atmosphere due to the re-entry of artificial space objects requires ongoing assessment (ESA-ESTEC 2024; Cirkovic 2025). These particles may be entirely new agents in different atmosphere layers or may exceed natural meteoroid influx for different metals under realistic future scenarios (Schulz and Glassmeier 2021). Given current projections, launch vehicles are likely to turn the space sector into a major contributor in various impact categories, including air acidification, climate change, and ozone depletion (Miraux, Wilson, and Dominguez Calabuig 2022).

The increasing number of rocket launches deposits water vapor and black carbon into the stratosphere, potentially affecting global temperatures and weather patterns (Ross and Scheaffer 2014). The energy used in space activities, if derived from fossil fuels, contributes to GHG emissions, exacerbating climate change.

Further research is necessary for understanding complex feedbacks between atmospheric layers and developing integrated models encompassing Earth system science and space physics (Lewis et al. 2011; Cnossen 2022; Laštovička et al. 2006; Laštovička 2013, 2023, 2024).

2.2 | Space Debris Challenges

Discarded satellites and rocket stages, debris from more than 600 fragmentation events (ESA 2023), slag and dust particles from more than 2400 solid rocket motor firings (ESA-MASTER 2020), coolant droplets from 16 reactor core ejection and two leakage events (ESA-MASTER 2020), as well as contributions from additional sources represent today's near-Earth space environment. It is the environment that operational spacecraft are exposed to during their mission. Potential collisions with debris lead to a range of consequences depending on the impact characteristics, including cratering and surface degradation effects, wall penetration, and local damage up to mission termination due to critical system failure or the complete disintegration of the satellite.

The current catalog of known space objects contains about 9500 active satellites and 19,000 debris objects (McDowell 2024; Space-Track 2024). About two-thirds of those objects reside in the low-earth orbit (LEO) region in altitudes between 200 km to 2000 km, where tracking primarily with ground-based radars comes with sensitivity limits on the order of a few cm up to about 10 cm. The European Meteoroid and Space debris Terrestrial Environment Reference (MASTER) model estimates that there are 36,500 objects larger than 10 cm, about 1 million objects larger than 1 cm, and more than 120 million objects larger than 1 mm (ESA 2023).

Given the currently assessed future growth of the space debris environment, combined with the drastic increase in launch activity over the past few years, and the increasing dependency on services from space, there is a substantial risk for increased satellite failure rates and potential service disruptions in the short-term future. Current adherence to post-mission disposal (PMD) practices required to avoid runaway growth scenarios is still too low, with only about 50% of naturally non-compliant satellites (or 70% for all satellites) in LEO complying, when the goal should be at least 90% for satellite constellations and even significantly higher.

The main approach toward disposal of spacecraft today and, in fact, being the one recommended by guidelines, as well as described as 'good practice' by many, is to foresee a re-entry into Earth's atmosphere by which the spacecraft would demise. It is well-known from past re-entry events that about 20% to 40% of the mass of a large spacecraft would not demise and reach the ground. Surviving fragments may pose a hazard to living communities on-ground, either through the kinetic impact energy or their potential release of hazardous substances (toxicity, radioactivity).

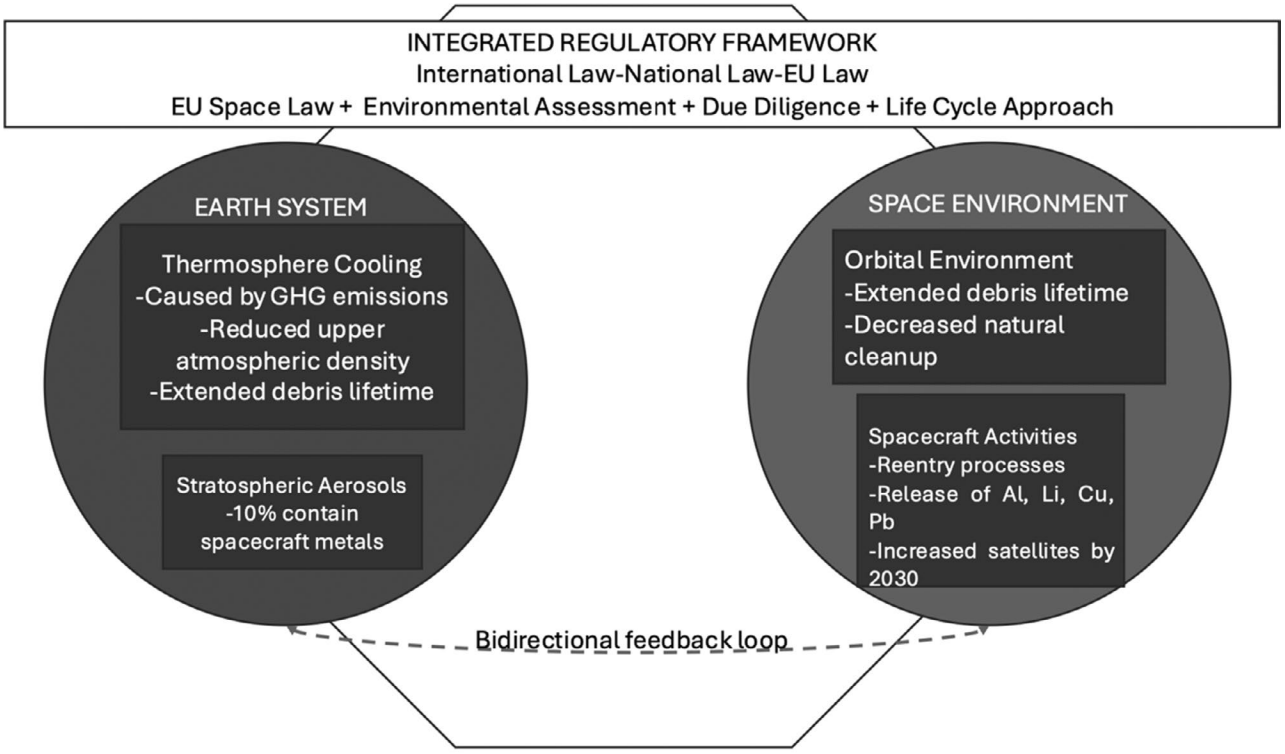
A well-established casualty risk threshold of 1 in 10,000, constrained to human population and human infrastructure, is typically evaluated in the design of a space mission to decide whether, after the end of the mission, it is allowed to undergo what is called an uncontrolled re-entry or, if exceeding the casualty risk threshold, measures need to be taken to target a controlled re-entry over an area with low human population density, a region generally identified as the so-called South-Pacific Ocean Uninhabited Area (SPOUA).

For the period between 2010 and 2020, about 100 tons of mass have re-entered Earth's atmosphere annually (ESA 2023). This resulted in an actual global casualty risk of 1 in 100 (or 1%) annually (Paradini and Anselmo 2022) for a period that can be considered a baseline to compare with the emerging trend of small satellites and large constellations. Current analyses considering those trends indicate that annual collision risk may increase to levels of 20% in the short term or even to 50% by 2050 (Pardini and Anselmo 2022). Considering the anticipated increase in re-entry rates and an associated rise in casualty risk, the space sector can be expected to address the question to which extent such risk is acceptable.

ESA's Clear Space 1 mission planned for 2028 and led first by Clear Space SA and then, since 2024, by OHB, aims to address various stages of the life cycle of space objects (ClearSpace 2024). For the first robotic mission to remove a piece of space debris from orbit, Clearspace-1 will rendezvous with, capture, and safely bring down a large derelict object for a safe atmospheric re-entry.

This mission is part of ESA's Space Safety Programme, which promotes the development of design guidelines and best practices for spacecraft that facilitate safe and controlled re-entry or disposal. This includes considerations such as the use of materials that minimize the generation of debris during re-entry and the inclusion of features that enable future active debris removal missions. During the operational phase, the program, especially via the

17585899, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1758-5899.70020 by New Aarhus University, Wiley Online Library on [27/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



© AUTHOR 2025. All rights reserved.

FIGURE 1 | Integrated Regulatory Framework for holistic space law approaches.

newly established Zero Debris policy, aims to mitigate the generation of new debris by promoting responsible operational practices and adherence to space debris mitigation guidelines. For the Zero Debris policy, but also internal ESA policies since 2014, the mitigation guidelines were transformed into mandatory requirements whenever ESA was responsible for a satellite mission.

2.3 | Life Cycle Assessment for Space Activities

This image visualizes how human activities create bidirectional environmental impacts between Earth's systems and outer space. It illustrates how greenhouse gas emissions cool the thermosphere, reducing atmospheric density and extending the orbital debris lifetime. Simultaneously, it shows how spacecraft reentry processes deposit metals in stratospheric aerosols, with 10% currently containing spacecraft metals. The diagram highlights the feedback loop where increasing space activities and emissions compound these effects over time. At the bottom, the integrated regulatory framework demonstrates how EU Space Law, environmental assessment, due diligence, and lifecycle approaches could address these interconnected challenges (Figure 1).

LCAs have been widely used to evaluate environmental impacts (Guinée and Lindeijer 2002; Benoît and Mazijn 2009). ESA's Clean Space initiative, particularly its EcoDesign strategy, has been at the forefront of quantifying the environmental impacts of the space sector using ISO-standardized life cycle assessment (LCA) methodologies (ISO 14040 and 14044). ESA has adapted these methodologies in a space-specific LCA Handbook published in 2017, accompanied by a dedicated LCA database. These tools support environmental assessment across ESA programs, including Copernicus and Ariane 6, and aim to guide the European space industry in applying LCA across spacecraft, launchers, propellants, and debris mitigation strategies.

Space mining, often referred to as 'space resource extraction' or 'astro-mining,' involves the extraction and utilization of resources from celestial bodies, such as asteroids, moons, and planets. Each term can refer to specific aspects or locations of resource extraction in space. The research on space technology's lifecycle and impact requires further investigation in relation to EPB. The potential environmental implications of space mining activities, both in space and on Earth, necessitate the development of comprehensive regulatory frameworks and assessment methodologies that integrate with existing EPB research to ensure sustainable development of this emerging industry.

3 | Future Generations, Due Diligence, and Strict Liability in Outer Space: General International Law

Due diligence requires states to consider the long-term impacts of their activities on the environment and resources, ensuring that their actions do not compromise the ability of future generations to meet their own needs (Foster 2021 a). As expressed in the Brundtland report, sustainable development is development that 'meets the needs of the present, without compromising the ability of future generations to meet their own needs' (World Commission on Environment and Development 1987, 27). The elements of sustainable development include integration of environmental protection and economic development, the right to

17585899, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1758-5899.70020 by New Aarhus University, Wiley Online Library on [27/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

development, sustainable utilization and conservation of natural resources, inter-generational equity, intra-generational equity, and procedural elements including environmental impact assessment, public participation in environmental decision-making, and access to information (Boyle et al. 2021).

3.1 | Historical Development of Intergenerational Equity Principles

The idea that environmental policies 'should be integrated with development planning' had featured centrally in the 1971 Founex report (UN, 1971). The 1972 Stockholm Declaration formally recognized a responsibility to future generations in Principle 1 and an integrated approach to development in Principle 13 (UN Doc A/CONF/48/14/REV1 1972). The Nordic compromise proposal for Principle 3 of the Rio Declaration stated the following: 'The right to development must be fulfilled so as to equitably meet developmental and environmental needs of present and future generations' (UN Doc A/CONF/151/26/Rev1 1992). Principle 3 has come to be considered 'the principal statement of the concept of intergenerational equity at international law' (Molinari 2015).

Specifically referencing intergenerational equity, the Institut de Droit International Resolution on 'Environment' (Strasbourg Session, 4 September 1997) states the following: 'Every State, when intervening on the basis of decisions taken in the exercise of its sovereignty in fields of activity where the effects of such decisions on the environment are clear, has the responsibility to ensure that activities within its jurisdiction or under its control do not cause damage which may affect the lives of the present and future generations'.

3.2 | Application to Atmosphere and Outer Space Activities

In the context of activities affecting the atmosphere, the International Law Commission's 2021 Draft Guidelines on the Protection of the Atmosphere state the following: 'The atmosphere should be utilized in an equitable and reasonable manner, taking fully into account the interests of present and future generations' (UN Doc A/76/10 2021, pp. 22–23). The commentary notes that this guideline addresses aspects of intra- and intergenerational equity. The phrase 'taking fully into account the interests of' was preferred to 'for the benefit of' present and future generations of humankind. The words 'the interests of', rather than 'the benefit of', were used to signal the integrated nature of the atmosphere, the 'exploitation' of which needs to consider a balancing of interests to ensure sustenance for the Earth's living organisms. The word 'fully' emphasizes the importance of taking various factors and considerations into account, and it should be read in conjunction with the seventh preambular paragraph, which recognizes that the interests of future generations in the long-term conservation of the atmosphere's quality should be fully considered (UN Doc A/76/10 2021, pp. 22–23).

Existing soft law principles of intergenerational equity and due diligence are reflected in various space-specific instruments, such as the Space Debris Mitigation Guidelines of the United Nations Committee on the Peaceful Uses of Outer Space (UN Doc A/62/20, Annex n.d.) and the Guidelines for the Long-term Sustainability of Outer Space Activities (UN Doc A/74/20, Annex II n.d.). These guidelines include limiting debris released during normal operations, minimizing the potential for on-orbit break-ups, post-mission disposal, and prevention of on-orbit collisions (IADC Space Debris Mitigation Guidelines, IADC-02-01, Rev. 1 2007).

Article II of the Registration Convention requires states to maintain a national registry of objects launched into outer space and provide information to the United Nations Secretary-General (Convention on Registration of Objects Launched into Outer Space 1975). States are encouraged to share relevant information and cooperate with other states and international organizations to promote the safe and responsible use of outer space, as outlined in the Declaration on International Cooperation in the Exploration and Use of Outer Space for the Benefit and in the Interest of All States, Taking into Particular Account the Needs of Developing Countries (A/RES/51/122 1996). Article 2 of the Declaration emphasizes the importance of international cooperation in the exploration and use of outer space for peaceful purposes and the need to consider the needs and interests of developing countries.

3.3 | Jurisdictional Issues and State Responsibility

The Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies (Outer Space Treaty, 1967) does not provide a clear definition of the extent of a state's jurisdiction in outer space. Instead, it establishes several key principles relating to jurisdiction and state responsibilities in conducting outer space activities. Article VI establishes that states bear international responsibility for national activities in outer space, regardless of whether such activities are carried out by governmental or non-governmental entities. Article VIII establishes that states maintain jurisdiction over their registered space objects and personnel. Article IX requires states to conduct their activities in outer space with due regard to the corresponding interests of other states and to avoid harmful contamination of celestial bodies and adverse changes in the Earth's environment.

Jurisdictional issues have been addressed in other regimes. For instance, in the context of the legal interests of future generations affected by the injurious use of states' territory and insufficient regulation of actors and activities under a state's jurisdiction and control, states must take fully into account future generations' interests alongside those of present generations. This argument also considers trends in the international legal articulation of sovereignty, as expressed particularly by the ICJ in Certain Activities Carried Out by Nicaragua in the Border Area (Costa Rica v Nicaragua) (ICJ 2015) ICJ Rep 665.

3.4 | Liability Regimes in Space Law

Liability for damage caused by space objects forms an important part of the due diligence building blocks in outer space law and

17585899, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1758-5899.70020 by New Aarhus University, Wiley Online Library on [27/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

activities. Indeed, outer space law includes strict liability, and in international law, strict liability regimes were negotiated and adopted to address the unique risks and potential for significant transboundary harm associated with these specific activities.

The Convention on International Liability for Damage Caused by Space Objects (Liability Convention, 1972) establishes a dual liability regime for damage caused by space objects, with states bearing absolute liability for damage on Earth's surface or to aircraft in flight and fault-based liability for damage to other space objects. Several international conventions have incorporated strict liability regimes, including the Vienna Convention on Civil Liability for Nuclear Damage (1963) and its 1997 Protocol and the International Convention on Civil Liability for Oil Pollution Damage (1969) and its 1992 Protocol.

The broader acceptance of strict liability as a general principle of international law remains unresolved and subject to ongoing debate. In outer space law, the Liability Convention reinforces the notion that states must exercise due diligence in supervising and controlling the space activities of both governmental and non-governmental entities under their jurisdiction, as mentioned in the International Law Commission's Draft Articles on Prevention of Transboundary Harm from Hazardous Activities (UN Doc A/56/10 2001).

3.5 | Due Diligence and Prevention of Harm

Due diligence obligations, strict liability, and due regard toward future generations are established in general international law (Foster 2021). As such, they are relevant to the ongoing interpretation and application of outer space law and have been included in various national space laws (UNOOSA, National Space Law database n.d.). Adherence to internationally recognized technical standards, guidelines, and best practices also entails states acting with due diligence toward preventing harmful interference and promoting the long-term sustainability of outer space activities. With continuous developments at different regulatory scales, particularly in the context of climate change, there are aspects that can be relevant for future developments in outer space law, generally, and EU Space Law, more particularly.

Article IX of the Outer Space Treaty reflects the obligation of every state not to allow knowingly its territory to be used for acts contrary to the rights and interests of others. This principle has been established in various international cases including the Trail Smelter Case (United States v Canada) (RIAA 1941) 3 RIAA 1905, the Island of Palmas Case (Netherlands v United States) (1928) 2 RIAA 829, and the Corfu Channel Case (United Kingdom v Albania) (1949) ICJ Rep 4. The effectiveness of due diligence obligations in preventing transboundary environmental harm and harm to common spaces and interests, such as outer space, will depend on their contextual application and the adoption of appropriate and sufficiently high levels of environmental protection.

The connection between climate change jurisprudence and outer space law can be established through the Outer Space Treaty (OST). OST Article I states that the exploration and use of outer space shall be carried out for the benefit and in the interests of all countries and shall be the province of all mankind. This could be read as indirectly supporting the concept of intergenerational equity by emphasizing the shared benefits of space activities for all of humanity. Similarly, Article IX requires states to conduct their space activities with due regard to the corresponding interests of other states, which can include individual state's definition of corresponding interests as including intergenerational equity and future generations and to avoid harmful contamination of celestial bodies and adverse changes in the Earth's environment. This provision reflects the principle of due diligence in preventing transboundary harm and protecting the outer space environment.

The customary international law principle of preventing significant harm to the environment of other states and areas beyond national jurisdiction raises the question of whether it should also encompass due regard for future generations in the context of climate change. The International Court of Justice (ICJ) has recognized the general obligation of states to ensure that activities within their jurisdiction and control respect the environment of other states and areas beyond national control (I.C.J. Reports 1996, 226). This obligation is further reinforced by the 'no-harm' rule and the maxim 'sic utere tuo ut alienum non laedas,' which govern how states must act with respect to the legal rights and interests of others.

Although jurisdiction in outer space is more complex due to its status as an area beyond national jurisdiction, the 'no-harm' rule and the maxim can still be applied to outer space law. These general principles of international law govern the conduct of states in relation to the rights and interests of other states and the international community as a whole and are specifically reflected in Articles VII and IX of the OST. As the exploration and use of outer space expands, it becomes increasingly important to ensure that these activities are conducted in a sustainable manner that does not compromise the ability of future generations to benefit from outer space.

The application of these principles to outer space activities provides a framework for addressing emerging challenges such as space debris management, resource utilization, and environmental protection beyond Earth's atmosphere. By integrating established principles of international environmental law with the specific context of outer space activities, the international community can develop more robust governance mechanisms that safeguard the interests of both present and future generations.

4 | EU Public-Private Interactions and Sustainability

4.1 | Environmental Competencies and Policy Objectives

The Treaty on the Functioning of the European Union (TFEU) establishes fundamental environmental competencies and policy objectives through Article 191. This primary law provides the legal basis for environmental protection, including the precautionary principle and measures addressing environmental challenges at regional and global levels (2012) OJ C326/47. The article emphasizes the 'polluter pays' principle and sets objectives for preserving, protecting, and improving environmental quality, while promoting measures to address regional and global environmental problems.

17585899, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1758-5899.70020 by New Aarhus University, Wiley Online Library on [27/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The Environmental Impact Assessment (EIA) Directive (Directive 2011/92/EU 2011) OJ L26/1, as amended by Directive 2014/52/EU [2014] OJ L124/1, requires assessment of environmental impacts before project approval. The Directive applies to major projects like nuclear power plants, railways, highways, waste disposal facilities, and dams over certain size thresholds. For other projects in areas like urban development, roads, and tourism, EU member states decide on EIA requirements case-by-case or through set criteria. While not directly targeting the space sector, projects necessary for space operations may fall under the Directive's scope.

The EIA process requires examination of direct and indirect effects on environmental factors, including population, biodiversity, land, soil, water, air, climate, landscape, material assets, and cultural heritage. Project developers must provide detailed information about potential impacts, alternatives considered, and mitigation measures. Public participation is mandatory, with requirements for informing the public about planned projects and providing opportunities for involvement in decision-making. EU member states report implementation data to the European Commission every six years, and the Commission periodically reviews the Directive's effectiveness.

In the context of space activities, the principles of due diligence and prevention of transboundary harm are implemented through national member state laws. The EU, as a signatory to the major international space treaties, is bound by these principles. The European Commission's Communication on European Space Policy COM (2007) 212 final emphasizes the importance of ensuring sustainable and peaceful use of outer space. The European Space Agency has established its framework for space activities through the Convention for the Establishment of a European Space Agency (n.d.); (adopted 30 May 1975, entered into force 30 October 1980) 1297 UNTS 161.

4.2 | Corporate Sustainability Due Diligence Directive

The European Parliament passed the Directive on Corporate Sustainability Due Diligence (EU) (Directive 2024/1760) (CSDDD 2024) on 24 April 2024. This directive is relevant to the space sector, as space activities and companies face significant environmental impacts and sustainability challenges. Space launches and spacecraft manufacturing contribute to greenhouse gas emissions, while space debris poses risks to the sustainability of orbital activities. The extraction and utilization of space resources raise concerns about environmental impact and equitable resource use.

The CSDDD applies to large companies based on specific size thresholds. For EU companies, the directive covers those with more than 500 employees and net worldwide turnover exceeding 150 million euros. A second tier includes companies with more than 250 employees and net worldwide turnover exceeding 40 million euros, provided at least 20 million euros is generated in high-impact sectors. Non-EU companies are subject to the directive if they generate more than 150 million euros turnover in the EU, with a lower threshold of 40 million euros applying if at least 20 million euros comes from high-impact sectors. While small and medium-sized enterprises (SMEs) are not directly subject to the CSDDD's obligations, they may be indirectly affected when they are part of larger companies' value chains. Companies meeting the thresholds must conduct due diligence across their operations, subsidiaries, and business relationships, addressing both human rights and environmental impacts, whether these occur within or outside the EU.

The financial sector faces modified obligations, with requirements applying to their own operations and upstream supply chains rather than full value chains. They must ensure that their business models align with Paris Agreement (2016) objectives (adopted 12 December 2015, entered into force 4 November 2016) OJ L282/4. While product disposal and indirect downstream business partners are excluded from the scope, all companies must adopt and implement climate transition plans aligned with the Paris Agreement.

EU member states have two years to transpose the directive into national legislation. The CSDDD introduces rules on administrative sanctions, liability in court, and civil liability, while affecting government procurement. It requires a risk-based approach aligned with frameworks such as the UN Guiding Principles on Business and Human Rights (UN Doc A/HRC/17/31 2011, 21 March 2011) and the OECD (2011) Guidelines for Multinational Enterprises.

The horizontal regulatory landscape also includes sector-specific requirements imposed by Regulation (EU) (2023)/1115 OJ L222/1 (Regulation on Deforestation-free Products), Regulation (EU) (2017)/821 OJ L130/1 (Conflict Minerals Regulation), Regulation (EU) (2023)/1542 [2023] OJ L240/1 (Batteries Regulation), and the Proposal for a Regulation on Prohibiting products made with forced labour (COM/2022/453 final). The CSDDD operates alongside Directive (EU) (2022)/2464 OJ L322/15 (Corporate Sustainability Reporting Directive) and Regulation (EU) (2019)/2088 OJ L317/1 (Sustainable Finance Disclosure Regulation).

Within this framework, the CSDDD established itself as the primary EU-wide due diligence regime, while explicitly stating that its obligations do not override more specific EU regulations. As implementation progresses across member states, further complications and potential reversals ensue, which we discuss in the following sections.

4.3 | Financial Regulations and Sustainable Activities

The EU Taxonomy for Sustainable Activities creates a systematic classification system for environmentally sustainable economic activities. Established through the Taxonomy Regulation of 2020, this framework aims to transform how sustainability is defined, measured, and incorporated into investment decisions across European financial markets. It addresses six environmental objectives encompassing climate change mitigation and adaptation, sustainable use of water resources, circular economy transition, pollution prevention, and biodiversity protection. The framework employs a three-part test to determine sustainability: activities must substantially contribute to at least

17585899, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1758-5899.70020 by New Aarhus University, Wiley Online Library on [27/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

one environmental objective, avoid significant harm to any other objectives, and meet minimum social safeguards. This approach ensures that activities classified as sustainable deliver genuine environmental benefits without creating offsetting harms in other domains.

The operational implementation of the Taxonomy occurs through Technical Screening Criteria established in various Delegated Acts. The Climate Delegated Act, adopted in 2021, covers activities contributing to climate objectives, while the Complementary Climate Delegated Act addresses specific nuclear and gas activities. These criteria establish specific thresholds and requirements for over 150 economic activities across diverse sectors including manufacturing, energy, transportation, buildings, and forestry. The "Do No Significant Harm" criteria are meant to ensure that activities making positive contributions in one area do not undermine other environmental objectives.

Companies subject to the Taxonomy Disclosure Delegated Act must report their alignment through key performance indicators. Non-financial companies report the proportion of their turnover, capital expenditure, and operating expenditure associated with Taxonomy-aligned activities. Financial institutions employ specialized indicators, including the green asset ratio for credit institutions, green investment ratio for asset managers, and specialized metrics for insurance underwriting.

The Taxonomy operates as the foundation of a broader sustainable finance ecosystem, connecting with the Sustainable Finance Disclosure Regulation, CSRD, and EU Green Bond Standard.

After nearly three years of implementation, practical experience has revealed several challenges requiring attention. The complexity of the "Do No Significant Harm" criteria has proven particularly difficult to apply, especially for backward-looking assessments of activities established before the criteria entered force. Data availability presents significant obstacles for financial institutions attempting to assess exposures to entities not subject to European reporting requirements. The reporting burden has proven substantial, especially for smaller entities with limited resources for compliance. The green asset ratio for credit institutions faces structural asymmetry issues that limit its effectiveness as a measurement tool.

Responding to these implementation challenges, the EU Platform on Sustainable Finance has developed recommendations focused on enhancing usability while preserving environmental integrity. These recommendations propose making the operating expenditure KPI voluntary except for research and development costs, introducing materiality thresholds aligned with financial reporting practices, ensuring symmetrical composition of the green asset ratio, simplifying assessment methodologies for retail exposures, and developing tailored approaches for small and medium-sized enterprises. The platform estimates that these changes would reduce reporting requirements by more than one-third compared to current standards while maintaining the framework's environmental ambition.

Despite implementation challenges, the taxonomy has demonstrated significant market impact. Green revenue across reporting entities increased by 22% between fiscal years 2022 and 2023, reaching €814 billion. Green capital expenditure grew by 32% during the same period, reaching €291 billion. Over €530 billion has been invested in key sectors including utilities, automotive, and transportation, demonstrating the framework's capacity to mobilize capital towards sustainability objectives. Beyond European markets, the taxonomy has established itself as a global benchmark, inspiring more than fifty-eight similar frameworks worldwide.

4.4 | Challenges to Sustainability Approaches in the EU

The EU approach to sustainability regulation increasingly emphasizes practical implementation alongside ambitious objectives. This calibration between environmental ambition and operational feasibility extends across the corporate sustainability due diligence directive and potential space governance frameworks. These domains recognize that effective regulation must reconcile transformative goals with implementation realities while acknowledging interconnections between financial markets, corporate operations, and emerging sectors like outer space.

President von der Leyen's mission letter to Commissioner-designate Andrius Kubilius positions space policy primarily within a security framework. Kubilius is tasked with overseeing "the continuous implementation of the EU Space Strategy for Security and Defence," working under the Executive Vice-President for Tech Sovereignty, Security and Democracy and in close collaboration with the High Representative/Vice-President. This organizational structure signals a security-oriented framing for space policy.

The letter specifically directs Kubilius to "lead the work on a future proposal for an EU Space Law" that will "introduce common EU standards and rules for space activities and harmonise licensing requirements." This instruction connects to recommendations in the Draghi Report on European competitiveness, suggesting an economic and industrial focus for the proposed legislation. Kubilius must also "propose a space data economy strategy to tap into the potential of space-derived data, products, and technology," highlighting economic dimensions of space policy.

The letter emphasizes industrial and economic objectives by instructing Kubilius to "foster a strong and innovative space industry" with "cost-effective access to space" and to develop EU space assets including Copernicus, Galileo, EGNOS, GOVSATCOM, and IRIS2. While environmental considerations are not specifically mentioned for space policy, von der Leyen establishes sustainability as a horizontal requirement, expecting all Commissioners to "contribute to achieving our agreed climate objectives" and ensure delivery of EU targets and sustainable development goals. However, these horizontal requirements remain unelaborated regarding space policy.

Von der Leyen emphasizes regulatory coherence and simplification, directing Commissioners to make rules "fit-for-purpose," reduce administrative burdens, and cut reporting obligations by at least 25% (35% for SMEs). The letter mandates stakeholder

17585899, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1758-5899.70020 by New Aarhus University, Wiley Online Library on [27/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

engagement through implementation dialogs, evidence-based policymaking supported by the Joint Research Centre, and youth policy dialogs. The letter does not address potential tensions between security, economic, and sustainability objectives in space governance.

The European Commission has proposed significant modifications to corporate sustainability reporting and due diligence requirements (COM(2025) 81 final, 2025/0045 (COD)) (2025a) to reduce administrative requirements while preserving European Green Deal objectives. The Corporate Sustainability Reporting Directive modifications in Articles 19a(1) and 29a(1) of Directive 2013/34/EU now restrict mandatory reporting to organizations with over 1000 employees, excluding approximately 80% of previously covered organizations. Companies must also exceed either €50 million turnover or €25 million balance sheet total to qualify as "large." Value chain reporting requirements have been limited through Articles 19a(3) and 29a(3), prohibiting companies from seeking information from entities with up to 1000 employees beyond voluntary reporting standards, except for information "commonly shared between undertakings in the sector concerned." Assurance providers must respect these limitations according to the new Article 34(2a).

The net turnover threshold for third-country undertakings has increased from €150 million to €450 million (Article 40a(1), fifth subparagraph) and for branches from €40 million to €50 million (Article 40a(1), fourth subparagraph), aligning the scope with CSDDD thresholds. The proposal eliminates Article 29b(1) requirements for sector-specific reporting standards and modifies Article 26a of Directive 2006/43/EC to streamline assurance requirements. Companies with more than 1000 employees but turnover under €450 million gain flexibility in reporting their taxonomy alignment under Regulation (EU) 2020/852, including reporting on partially aligned activities (Articles 19b and 29aa).

The CSDDD undergoes substantial transformation through Article 8 amendments that focus due diligence primarily on direct business partners, with indirect partner investigation required only when "plausible information" suggests adverse impacts, defined in Recital 21 as "information of an objective character that allows the company to conclude that there is a reasonable likelihood that the information is true." Monitoring assessments now occur "at least every 5 years" rather than annually per the revised Article 15. Article 3(1)(n) narrows the stakeholder definition to directly affected parties, while Article 29 amendments remove the EU-wide liability regime in favor of national frameworks. Article 27(4) modifications remove the requirement to base penalties on worldwide turnover. Article 4 extends maximum harmonization provisions to core due diligence duties (Articles 6 and 8, Article 10(1) to (5), Article 11(1) to (6), and Article 14), preventing member states from creating stricter requirements in these areas.

The obligation to terminate business relationships with severe impacts has been replaced with a suspension requirement under Articles 10(6) and 11(7), providing more flexibility for companies working with suppliers to address issues. Despite streamlining, the proposal maintains environmental commitments, modifying Article 22(1) of the CSDDD to require companies to develop climate transition plans "including implementing actions" aligned with the Paris Agreement's 1.5°C warming limit.

For implementation, member states have 12 months to transpose this directive after entry into force (Article 5). The Commission will adopt voluntary reporting standards within 4 months (Article 29ca) and revise the ESRS delegated act within 6 months. The deadline for general due diligence guidelines has been advanced to July 2026 (Article 19(3)), while a separate proposal seeks to delay implementation deadlines by 1–2 years. Technical clarifications include exempting the European Financial Stability Facility from sustainability reporting requirements (Article 1(4)) and confirming that companies "shall not be required to markup their sustainability reporting" until specific rules are adopted (Article 29d).

The proposal relies on Articles 50 and 114 of the Treaty on the Functioning of the European Union as its legal foundation, emphasizing market integration rather than environmental provisions in Articles 191–193 TFEU. This raises questions about alignment with the "high level of protection" standard established in Article 191(2) TFEU. The maximum harmonization provisions could restrict Member States' ability to implement stronger environmental measures under Article 193 TFEU, while the focus on direct business partners may weaken preventive action throughout global value chains.

The European Commission's simplification of corporate sustainability requirements reveals patterns likely to influence EU Space Law development. Both frameworks balance ambitious objectives with implementation concerns, focus on economic competitiveness by referencing the Draghi Report, emphasize harmonization and standardization, and adopt stakeholder dialog approaches. These patterns suggest that space sustainability frameworks may formally recognize environmental considerations while subordinating them to security and economic priorities, adopt threshold-based approaches that exempt smaller actors, create tensions with TFEU environmental principles, rely heavily on voluntary measures, and face challenges translating objectives into enforceable regulations.

The developments in corporate sustainability regulation indicate that future EU Space Law will likely formalize sustainability requirements while limiting administrative burdens to maintain European competitiveness in space. This approach produces frameworks that appear comprehensive but may prove limited in practical application, particularly for smaller space actors. The EU regulatory philosophy attempts to maintain environmental ambition while addressing competitiveness concerns through implementation simplifications, inevitably creating trade-offs between comprehensive regulation and practical feasibility.

5 | Conclusions: Law, Politics, and Sustainability Governance

The European Commission's approach to corporate sustainability frameworks and space governance reveals significant tensions in the relationship between law, politics, and sustainability regulation. The recent modifications to the Corporate

17585899, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1758-5899.70020 by New Aarhus University, Wiley Online Library on [27/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Sustainability Reporting Directive and Corporate Sustainability Due Diligence Directive (COM(2025) 81 final) exemplify this tension—while maintaining formal commitment to environmental objectives, they substantially narrow practical application through increased employee thresholds (from 500 to 1000) and simplified implementation requirements.

This recalibration demonstrates how sustainability frameworks respond to political imperatives, particularly competitiveness concerns articulated in the Draghi Report. A similar pattern extends to space governance, where Commissioner-designate Kubilius's mandate simultaneously acknowledges sustainability as a horizontal requirement while emphasizing security and economic priorities. The adjustments to both frameworks reveal that established environmental standards remain susceptible to modification when confronted with economic pressures.

The malleability of sustainability law is evident in how the CSRD and CSDDD modifications maintain environmental commitments in principle while significantly reducing their scope through threshold adjustments. This pattern of "threshold politics" creates a disconnect between formal environmental objectives and their practical implementation, raising questions about the resilience of sustainability commitments when economic considerations become paramount.

Current developments indicate a shift from substantive to procedural environmental protection. The simplified CSRD and CSDDD frameworks place greater emphasis on voluntary standards and transparency mechanisms rather than comprehensive mandatory requirements. This procedural turn suggests recognition of the limitations of command-and-control regulation when balancing sustainability against competitiveness.

The regulatory landscape for space activities mirrors these challenges, operating through both horizontal dimensions (international space law, environmental law) and vertical dimensions (EU directives and national legislation). While existing EU frameworks like the EIA Directive could address environmental impacts of space activities, the recent patterns of regulatory simplification suggest that future space governance may similarly prioritize economic considerations over comprehensive environmental protection.

As member states implement the modified CSDDD and develop space laws, the fundamental challenge remains balancing effective environmental protection with economic competitiveness. The tension between maintaining high sustainability standards while reducing administrative burden creates uncertainty about the future effectiveness of EU climate governance across multiple domains. The long-term impact will depend on whether these simplified frameworks can deliver meaningful environmental protection while accommodating economic priorities—a balance that recent developments suggest remains elusive.

Acknowledgements

The authors would like to thank the anonymous reviewers for their constructive feedback. We also acknowledge the support of the Aarhus University for providing resources that contributed to this research.

Conflicts of Interest

The authors declare no conflicts of interest. While Vitali Braun provides consultancy services to the European Space Agency, the views expressed in this article are solely those of the authors and do not necessarily represent the views of ESA or any affiliated organization.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

References

A/RES/51/122. 1996. "Declaration on International Cooperation in the Exploration and Use of Outer Space for the Benefit and in the Interest of All States, Taking Into Particular Account the Needs of Developing Countries."

Austin, J. H., T. S. Soares, and L. Innocenti. 2015. "Developing a Standardised Methodology for Space-Specific Life Cycle Assessment, Challenges in European Aerospace." 5th CEAS Air & Space Conference (2015).

Benoît, C., and B. Mazijn. 2009. *Guidelines for Social Life Cycle Assessment of Products*. United Nations Environment Programme.

Boyle, A., C. Redgwell, and P. Birnie. 2021. *International Law and the Environment*. Oxford University Press.

Cirkovic, E. 2025. *The Law of Complex Earth and Outer Space Systems: The Cosmolegal Proposal*. Routledge.

ClearSpace. 2024. https://clearspace.today/.

Cnossen, I. 2022. "A Realistic Projection of Climate Change in the Upper Atmosphere Into the 21st Century." *Geophysical Research Letters* 49, no. 19: e2022GL100693.

Convention for the Establishment of a European Space Agency. n.d. "Convention for the Establishment of a European Space Agency Adopted 30 May 1975, Entered Into Force 30 October 1980 1297."

Convention on International Liability. 1972. "Convention on International Liability for Damage Caused by Space Objects."

Convention on Registration of Objects Launched Into Outer Space. 1975. "Convention on Registration of Objects Launched Into Outer Space 1023 UNTS 15."

CSDDD. 2024. "Directive on Corporate Sustainability due Diligence (EU) (Directive 2024/1760)."

Directive (EU) 2022/2464. 2022. "Corporate Sustainability Reporting Directive."

Directive 2011/92/EU 2011 OJ L26/1. 2011. "As Amended by Directive 2014/52/EU 2014 OJ L124/1 (Environmental Impact Assessment Directive)."

ESA. 2023. "ESA's Annual Space Environment Report." https://www.esa.int/Safety_Security.

ESA. 2024. "Space Safety Programme." https://www.esa.int/Space_Safety.

ESTEC. 2024. *Proceedings, Understanding the Atmospheric Effects of Spacecraft Re-entry Conference Proceedings, 10–11 Jan 2024*. ESTEC, Europe/Amsterdam, 2024.

ESA-MASTER. 2020. "Meteoroid and Space Debris Terrestrial Environment Reference." https://sdup.esoc.esa.int/.

European Commission. 2007. "Communication on European Space Policy COM(2007) 212 Final."

European Commission. 2022. "Proposal for a Regulation on Prohibiting Products Made With Forced Labour (COM/2022/453 Final)."

17585899, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1758-5899.70020 by New Aarhus University, Wiley Online Library on [27/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

European Commission. 2025a. "Proposal for a Directive of the European Parliament and of the Council Amending Directives 2006/43/EC, 2013/34/EU, (EU) 2022/2464 and (EU) 2024/1760 as Regards Certain Corporate Sustainability Reporting and due Diligence Requirements (COM(2025) 81 Final, 2025/0045 (COD))."

European Commission. 2025b. "Proposal for a Directive of the European Parliament and of the Council Amending Directives (EU) 2022/2464 and (EU) 2024/1760 as Regards the Dates From Which Member States are to Apply Certain Corporate Sustainability Reporting and due Diligence Requirements (COM(2025) 80 Final, 2025/0044 (COD))."

European Commission. 2025c. "Commission Staff Working Document Accompanying the Documents Proposal for a Directive of the European Parliament and of the Council Amending Directives 2006/43/EC, 2013/34/EU, (EU) 2022/2464 and (EU) 2024/1760 as Regards Certain Corporate Sustainability Reporting and due Diligence Requirements and Proposal for a Directive of the European Parliament and of the Council Amending Directives (EU) 2022/2464 and (EU) 2024/1760 as Regards the Dates From Which Member States are to Apply Certain Corporate Sustainability Reporting and due Diligence Requirements and Specifying Further Steps (SWD(2025) 80 Final)."

Foster, C. E. 2021. *Intergenerational Equity and Rights of Future Generations. Max Planck Encyclopedia of Public International Law*. Oxford University Press.

Guinée, J. B., and E. Lindeijer. 2002. *Handbook on Life Cycle Assessment: Operational Guide to the ISO Standards*. Springer.

I.C.J. Reports. 1996. "Legality of the Threat or use of Nuclear Weapons." Advisory Opinion, p. 226.

I.C.J. Reports. 2015. "Certain Activities Carried out by Nicaragua in the Border Area (Costa Rica v Nicaragua), p. 665."

IADC. 2007. "IADC Space Debris Mitigation Guidelines." IADC-02-01, Rev. 1.

Institut de Droit International. 1997. "Resolution on 'Environment'." Strasbourg Session, 4 September 1997.

International Convention on Civil Liability. 1969. "International Convention on Civil Liability for Oil Pollution Damage and its 1992 Protocol."

International Organization for Standardization. 2006. "ISO 14040 Environmental Management—Life Cycle Assessment—Principles and Framework." Geneva: International Organization for Standardization.

RIAA. 1928. "Island of Palmas Case (Netherlands v United States) 2 RIAA 829."

ISO. 2006. "ISO 14044 Environmental Management—Life Cycle Assessment—Requirements and Guidelines." Geneva: International Organization for Standardization.

Laštovička, J. 2013. "Trends in the Upper Atmosphere and Ionosphere: Recent Progress." *Journal of Geophysical Research: Space Physics* 118, no. 5: 3924–3935. https://doi.org/10.1002/jgra.50341.

Laštovička, J. 2023. "Progress in Investigating Long-Term Trends in the Mesosphere, Thermosphere, and Ionosphere." *Atmospheric Chemistry and Physics* 23: 5783–5800. https://doi.org/10.5194/acp-23-5783-2023.

Laštovička, J. 2024. "Dependence of Long-Term Trends in fof2 at Middle Latitudes on Different Solar Activity Proxies." *Advances in Space Research* 73: 685–689. https://doi.org/10.1016/j.asr.2023.09.047.

Laštovička, J., R. A. Akmaev, G. Beig, J. Bremer, and J. T. Emmert. 2006. "Global Change in the Upper Atmosphere." *Science* 314, no. 5803: 1253–1254.

Lewis, H. G., A. Saunders, G. Swinerd, and R. J. Newland. 2011. "Effect of Thermospheric Contraction on Remediation of the Near-Earth Space Debris Environment." *Journal of Geophysical Research* 116: A00H08.

McDowell, J. 2024. "General Catalog of Artificial Space Objects." https://planet4589.org/space/gcat/.

Miraux, L., A. R. Wilson, and G. J. Dominguez Calabuig. 2022. "Environmental Sustainability of Future Proposed Space Activities." *Acta Astronautica* 200: 329–346.

Molinari, C. 2015. "Principle 3: From a Right to Development to Intergenerational Equity." In *The Rio Declaration on Environment and Development: A Commentary*, edited by J. E. Viñuales, 139–156. Oxford University Press.

Murphy, D. M., M. Abou-Ghanem, D. J. Cziczo, et al. 2023. "Metals From Spacecraft Reentry in Stratospheric Aerosol Particles." *Proceedings of the National Academy of Sciences of the United States of America* 120, no. 43: e2313374120.

OECD. 2011. "Guidelines for Multinational Enterprises."

Pardini, C., and L. Anselmo. 2022. "Using the Space Debris Flux to Assess the Criticality of the Environment in Low Earth Orbit." *Acta Astronautica* 198: 756–760.

Paris Agreement. 2016. "Paris Agreement (Adopted 12 December 2015, Entered Into Force 4 November 2016) OJ L282/4."

Parker, W. E., M. K. Brown, and R. Linares. 2025. "Greenhouse Gases Reduce the Satellite Carrying Capacity of Low Earth Orbit." *Nature Sustainability*. https://doi.org/10.1038/s41893-025-01512-0.

Persson, L., B. M. Carney, A. Fairbrass, et al. 2022. "Outside the Safe Operating Space of the Planetary Boundary for Novel Entities." *Environmental Science & Technology* 56, no. 3: 1510–1521.

Regulation (EU). 2010. "Regulation (EU) No 911/2010 of the European Parliament and of the Council of 22 September 2010 on the European Earth Monitoring Programme (GMES) OJ L276/1."

Regulation (EU). 2013. "Regulation (EU) No 1285/2013 of the European Parliament and of the Council of 11 December 2013 on the Implementation and Exploitation of European Satellite Navigation Systems OJ L347/1."

Regulation (EU). 2014. "Regulation (EU) No 377/2014 of the European Parliament and of the Council of 3 April 2014 Establishing the Copernicus Programme OJ L122/44."

Regulation (EU). 2023. "Regulation (EU) 2023/588 of the European Parliament and of the Council of 15 March 2023 Establishing the Union Space Programme and the European Union Agency for the Space Programme OJ L80/1."

Regulation (EU). 2017. "Regulation (EU) 2017/821 OJ L130/1 (Conflict Minerals Regulation)."

Regulation (EU) 2020/852. 2020. "Taxonomy Regulation."

Regulation (EU) 2023/1115 OJ L222/1. 2023. "Regulation on Deforestation-Free Products."

Regulation (EU) 2023/1542 OJ L240/1. 2023. "Batteries Regulation."

Regulation (EU)/2088 2019. 2019. "Regulation (EU)/2088 2019OJ L317/1 (Sustainable Finance Disclosure Regulation)."

RIAA. 1941. "Trail Smelter Case (United States v Canada) 3 RIAA 1905."

Richardson, K., W. Steffen, W. Lucht, et al. 2023. "Earth Beyond Six of Nine Planetary Boundaries." *Science Advances* 9, no. 37: 1–19.

Ross, M., and P. Scheaffer. 2014. "Radiative Forcing Caused by Rocket Engine Emissions." *Earth's Future* 2, no. 4: 177–196.

Schulz, L., and K. H. Glassmeier. 2021. "On the Anthropogenic and Natural Injection of Matter into Earth's Atmosphere." *Advances in Space Research* 67, no. 2: 846–856.

Space-Track. 2024. "Space-Track.org." https://www.space-track.org/.

ICJ. 1949. "The Corfu Channel Case (United Kingdom v Albania) ICJ Rep 4."

17585899, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1758-5899.70020 by New Aarhus University, Wiley Online Library on [27/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TFEU. 2012. "The Treaty on the Functioning of the European Union (TFEU) OJ C326/47."

Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space. 1967. "Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies (Outer Space Treaty, 1967) 610 UNTS 205."

UN Doc A/56/10. 2001. "Draft Articles on Prevention of Transboundary Harm From Hazardous Activities."

UN Doc A/62/20, Annex. n.d. "Space Debris Mitigation Guidelines of the United Nations Committee on the Peaceful Uses of Outer Space."

UN Doc A/74/20, Annex II. n.d. "Guidelines for the Long-Term Sustainability of Outer Space Activities."

UN Doc A/76/10. 2021. "Draft Guidelines on the Protection of the Atmosphere."

UN Doc A/CONF.151/26/Rev.1. 1992. "Rio Declaration on Environment and Development."

UN Doc A/CONF/48/14/REV.1. 1972. "Declaration of the United Nations Conference on the Human Environment."

UN Doc A/HRC/17/31. 2011. "UN Doc A/HRC/17/31 UN Guiding Principles on Business and Human Rights."

UN Doc A/RES/51/122. 1996. "Declaration on International Cooperation in the Exploration and Use of Outer Space for the Benefit and in the Interest of All States, Taking Into Particular Account the Needs of Developing Countries."

UN Office for Outer Space Affairs (UNOOSA). National Space Law Database n.d. https://www.unoosa.org/oosa/en/ourwork/spacelaw/nationalspacelaw/index.html.

United Nations. 1971. "Development and Environment: Report and Working Papers of a Panel of Experts Convened by the Secretary-General of the United Nations Conference on the Human Environment (The Founex Report)." Founex: United Nations.

Vienna Convention on Civil Liability. 1963. "Vienna Convention on Civil Liability for Nuclear Damage and its 1997 Protocol."

World Commission on Environment and Development. 1987. *Our Common Future*. Oxford University Press.

17585899, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1758-5899.70020 by New Aarhus University, Wiley Online Library on [27/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License